

Frontend Debugging Notes

Mistakes & the concepts behind them — What's For Dinner project

1. box-shadow not visible under the navbar

Mistake:

A box-shadow was added to the <nav> but never appeared on screen, even though the CSS looked correct.

Concept:

box-shadow paints outside an element's box without affecting layout. If a sibling element comes right after in the DOM and both sit in normal document flow, that sibling can visually sit on top of the shadow since stacking order (not just source order) decides what paints over what. Giving the shadow-casting element its own stacking context fixes this.

```
nav {  
  position: relative;  
  z-index: 10;  
  box-shadow: 0px 8px 10px -4px rgba(0,0,0,0.3);  
}
```

2. Profile text wrapping under the avatar image instead of beside it

Mistake:

The '.image-icon' wrapper had a fixed width/height (40x40px), and the inside was set to width:100%, leaving zero horizontal space for the 'Profile' text next to it.

Concept:

When a fixed-size wrapper's child fills 100% of that fixed size, there's no room left for anything else inside it to sit beside it. The fixed size should live on the element that actually needs a fixed shape (the image itself), while the wrapper stays flexible so it can grow to fit both the image and the text.

```
/* fix: put the fixed size on the img, not the wrapper */  
.image-icon {  
  display: inline-flex;  
  align-items: center;  
  gap: 5px;  
}  
.image-icon img {  
  width: 40px;  
  height: 40px;  
  border-radius: 50%;  
}
```

3. Scrollbar overflowing outside a card's rounded corners

Mistake:

A scrollable div also had border-radius directly on it, so the scrollbar rendered as a straight edge poking past the rounded corner.

Concept:

A single element can't clip its own scrollbar to its own border-radius reliably. The fix is to separate responsibilities: an outer wrapper holds 'overflow:hidden' + the border-radius (so it visually clips anything inside, including the scrollbar), while an inner element holds 'overflow-y:auto' and does the actual scrolling.

```
.card { border-radius: 16px; overflow: hidden; }
.card-scroll { overflow-y: auto; max-height: 500px; }
```

4. Custom scrollbar color / hover state

Mistake:

Wanted to change the scrollbar's default gray color, and its color on hover.

Concept:

Scrollbar styling is not fully standardized across browsers. Firefox uses the 'scrollbar-color' + 'scrollbar-width' properties (color only, no hover control in pure CSS). Chrome/Edge/Safari use the '::-webkit-scrollbar' family of pseudo-elements, which DO support ':hover' on the thumb. Both need to be written together to cover all browsers; Firefox hover requires JS if it's ever needed.

```
scrollbar-width: thin;
scrollbar-color: #FFDBD2 white;

&::-webkit-scrollbar-thumb { background: #FFDBD2; }
&::-webkit-scrollbar-thumb:hover { background: #FF5D00; }
```

5. Cannot read properties of undefined (reading 'rate')

Mistake:

showMeal() tried to read Meals[mealIndex].rate, but mealIndex was declared and never assigned before use.

Concept:

A variable that is declared with 'var' but never assigned is 'undefined'. Using it as an array index gives Meals[undefined], which returns undefined — and reading any property off undefined throws a TypeError. The function that generates the random index (makeRandomNumber) existed but was never actually called before showMeal() needed the value.

```
function showMeal(){
  makeRandomNumber(); // must run first
  // ...now Meals[mealIndex] is valid
}
```

6. innerHTML() called as a function

Mistake:

Code was written as element.innerHTML(mealImageDiv) with parentheses, which throws 'innerHTML is not a function'.

Concept:

innerHTML is a property of a DOM element, not a method — it must be assigned to with '=', not invoked with '()'. This is a common mix-up coming from methods like .append() or .insertAdjacentHTML() which ARE functions.

```
element.innerHTML = mealImageDiv; // correct
element.innerHTML(mealImageDiv); // wrong — throws TypeError
```

7. Whole card content disappearing after updating one section

Mistake:

Setting '#meals-div'.innerHTML = mealImageDiv wiped out the entire card (ingredients, instructions, nutrition, everything), leaving only the small image block that was assigned.

Concept:

innerHTML = replaces ALL existing children of that element, not just adds to them. If a parent container holds multiple sections but only one small piece of HTML is being generated in JS, assigning it directly to the shared parent destroys every sibling section that lived there. The fix is to give each independently updated piece (image, ingredients, instructions, nutrition, chef tips) its own dedicated container id, and update each one individually — matching the id-per-section structure already set up in index.html.

```
// each section gets its own target, not one shared parent
document.getElementById('meal-image').innerHTML = mealImageDiv;
document.getElementById('ingredients-div').innerHTML = ingredientsHTML;
document.getElementById('instructions-div').innerHTML = instructionsHTML;
```

8. col-xs-4 not working on small screens (Bootstrap version mismatch)

Mistake:

A grid column used 'col-lg-4 col-md-4 col-sm-4 col-xs-4', expecting col-xs-4 to control the smallest screens. Below 576px, the layout broke and columns stacked full-width instead of staying in a 3-column row.

Concept:

Bootstrap 5 removed the 'xs' breakpoint prefix entirely. The smallest/base tier in Bootstrap 5 is written WITHOUT any breakpoint prefix at all — just 'col-4' — since that unprefixed class already applies from 0px up by default. 'col-xs-4' is a Bootstrap 3/4 class name; in Bootstrap 5 it's simply not a recognized class and gets silently ignored, so nothing applies below the 'sm' breakpoint.

```
/* Bootstrap 5 – no xs prefix, base tier is unprefixed */
col-4 /* applies from 0px up (was col-xs-4) */
col-sm-4 col-md-4 col-lg-4 /* only needed if the value changes per breakpoint */
```

Compiled from the 'What's For Dinner' project debugging session. Keep this file in the project root and review it before starting new work.